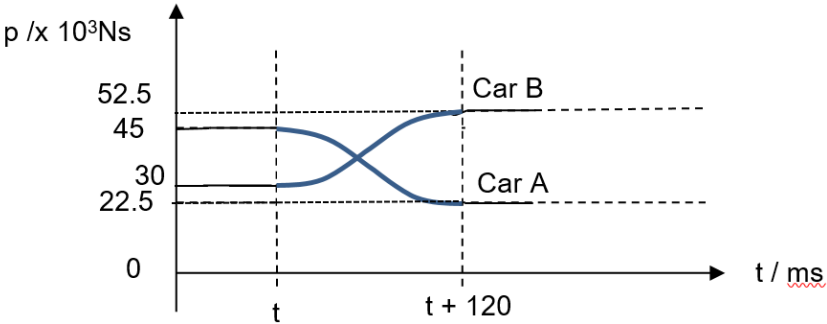


|                |                                                                                                                                                                                                                                                                                                                                                                         |                                                                                        |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>3(a)(i)</b> | Using Principle of Conservation of Linear Momentum<br>$900 \times 50 + 1500 \times 20 = 900 \times 25 + 1500 \times v_B$<br>$v_B = 35 \text{ m s}^{-1}$                                                                                                                                                                                                                 | <b>[1] ans</b>                                                                         |
| <b>(ii)</b>    |  <p><b>[1]</b> Position and values of initial (A&amp;B) and final (A only) momentum shown correctly.</p> <p><b>[1]</b> Change of momentum takes place over same time interval.</p> <p><b>[1]</b> Final momentum of B deduced correctly. (credit given if ans is given in (a)(ii))</p> | <b>[1] correct shape</b><br>i.e. graphs between $t$ and $(t+120)$ are curved correctly |
| <b>(b)</b>     | Apply $F = \frac{\Delta p}{\Delta t} = \frac{(22.5 - 45) \times 10^3}{120 \times 10^{-3}} = -1.9 \times 10^5 \text{ N}$                                                                                                                                                                                                                                                 | <b>[1] ans</b>                                                                         |
| <b>(c)</b>     | By undergoing deformation, <b>seat belts / air bags / crumple zones increase the time</b> for the body to be <b>brought to rest / change momentum</b> .<br>According to Newton's second law of motion, as <b>change of momentum is the same</b> , increase in time will reduce the resultant force acting on the body.                                                  | <b>[2]</b>                                                                             |
| <b>MC</b>      | accept: same impulse, some energy converted to elastic PE of seat belt.<br>Feature + longer time <b>[1]</b> ; feature + same change in momentum <b>[1]</b>                                                                                                                                                                                                              |                                                                                        |

4(c)

Simple harmonic motion is defined as a periodic motion of a particle whose

[1]